

### 3.3.3 The Quotient Rule

**Recall:** Last class we learned that the *Product Rule* states:  $\frac{d}{dx}[f(x)g(x)] = f(x)g'(x) + g(x)f'(x)$

We will now address the problem of differentiating the quotient of two functions.

**Example 1:** Use the definition of the derivative to find a formula for  $\frac{d}{dx}\left[\frac{f(x)}{g(x)}\right]$  assuming that both  $f$  and  $g$  are differentiable and  $g(x) \neq 0$ .

The above examples provides proof of the following theorem.

**Theorem (The Quotient Rule):** If  $f$  and  $g$  are differentiable, then

*Note: The formula for the derivative produced by the quotient rule is valid for all  $x$  such that  $g(x) \neq 0$ .*

**Example 2:** If  $g(x) = \frac{2e^x}{x^2}$ , then find  $g'(x)$ .

**Example 3:** Find the first derivative of  $f(x) = \frac{2-x}{x^2+6}$ .

**Example 4:** Find an equation of the tangent line to the curve  $y = \frac{e^x}{1+x^2}$  when  $x = 1$ .

**Summary of derivative formulas we know so far:**

Assume that  $f$  and  $g$  are differentiable and  $c$  is a constant. We have:

- $\frac{d}{dx}[c] = 0$
- $(f \pm g)' = f' \pm g'$
- $\frac{d}{dx}[x^n] = nx^{n-1}$
- $\frac{d}{dx}[e^x] = e^x$
- $(cf)' = cf'$
- $(fg)' = fg' + gf'$
- $\left(\frac{f}{g}\right)' = \frac{gf' - fg'}{g^2}$  when  $g \neq 0$